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Table 13.2: Differences between Photosystem I and Photosystem II	
Photosystem I	**Photosystem II**
1. The reaction centre is P700	1. Reaction centre is P680
2. PS I is involved in both cyclic and non-cyclic.	2. PS II participates in Non-cyclic pathway
3. Not involved in photolysis of water and evolution of oxygen	3, Photolysis of water and evolution of oxygen take place.
4. It receives electrons from PS II during non-cyclic photophosphorylation	4. It receives electrons by photolysis of water
5. Located in unstacked region granum facing chloroplast stroma	5. Located in stacked region of thylakoid membrane facing lumen of thylakoid.
6. Chlorophyll and Carotenoid ratio is 20 to 30:1	6. Chlorophyll and Carotenoid ratio is 3 to 7:1

- PS II – CC II consists of reaction centre P680 and LHC II (Table 13.2).
- Light Harvesting Complex consists of several chlorophylls, carotenoids and xanthophyll molecules.
- The main function of LHC is to harvest light energy and transfer it to their respective reaction centre.

13.10 Photo chemical phase of light reaction

In this phase electrons pass through electron carrier molecules and generate assimilatory powers ATP and NADPH + H^+. Splitting of water molecule generates electrons replacing electrons produced by the light.

13.10.1 Photolysis of Water

The process of Photolysis is associated with **Oxygen Evolving Complex** (OEC) or water splitting complex in pigment system II and is catalysed by the presence of Mn^{++} and Cl^-. When the pigment system II is active it receives light and the water molecule splits into OH^- ions and H^+ ions. The OH^- ions unite to form water molecules again and release O_2 and electrons. Photolysis of water is due to strong oxidant which is yet unknown and designated as Z or Yz. Widely accepted theory proposed by **Kok** *et al.*,(1970) explaining photo-oxidation of water is **water oxidising clock** (or) **S' State Mechanism.** It consists of a series of 5 states called as S_0, S_1, S_2, S_3 and S_4. Each state acquires positive charge by a photon (*hv*) and after the S_4 state it acquires 4 positive charges, four electrons and evolution of oxygen. Two molecules of water go back to the S_0. At the end of photolysis 4 H^+,4 e^- and O_2 are evolved from water (Figure 13.13).

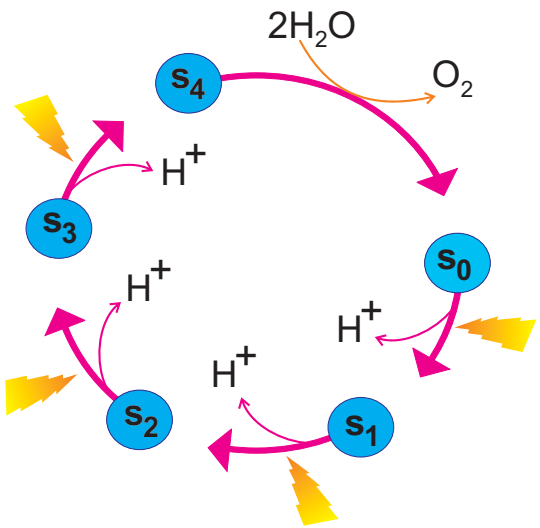


Figure 13.13: Oxygen Evolving Complex (OEC)